

# Quantum Gravitation

## Problem Set 5

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Topic

Massive Dispersion

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

### Problems

1. Derive the first displayed relation used in Massive Dispersion. State all assumptions and conventions.

$$E^2 = p^2 c_m^2 + m_g^2 c_m^4$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$v_g/c_m \approx 1 - m_g^2 c_m^4 / (2E^2)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$m_g \lesssim 10^{-32} \text{eV}/c^2$$

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Show enough intermediate work that another student can reproduce the calculation.